PROBLEM 43

Two Connected Particles through a Hole

Original statement · Problems.pdf, p. 30

Angular Momentum + Energy

We consider a smooth horizontal plane with a small hole in it. One of the ends of a massless cord of length $2a$ is passed through the hole and then, two point particles, P_1 and P_2, each with the same mass m, are attached to the ends of the cord. Therefore, P_1 is constrained to move on the plane and P_2 moves along a vertical line below the hole.

Initially, half of the cord lies under the plane and half of the cord is on the plane (so $r(0) = a$). At that instant, P_1 is pushed with a speed v_0 perpendicular to the cord, counterclockwise as seen from above. Assume that the cord is always in tension.

(a) What is the value of v_0 if P_1 describes a circumference?

(b) In that case, what is the value of the tension in the cord?

(c) If $v_0 = \sqrt{8ag/3}$, reduce the problem to integrals (i.e., quadratures).

(d) Show that for this value of v_0, P_1 oscillates between $r = a$ and $r = 2a$.

(e) Apply Newton's law to P_2 and show that the tension of the cord is $T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$.

(f) Write down the new equations of motion if the plane on which P_1 moves is rough, with a friction coefficient μ.

Hint: the roots of the polynomial $-3x^3 + 7x^2 - 4$ are $x_1 = 1$, $x_2 = 2$ and $x_3 = -2/3$.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** P_1 on a smooth horizontal plane at polar position (r, θ) and P_2 hanging on the vertical through the hole O, joined by an inextensible massless cord of length $2a$ passing through O.
- **Constraints and degrees of freedom:** the cord length gives $r + \ell_2 = 2a$, where ℓ_2 is the hanging length, so $z_2 = -\ell_2 = r - 2a$ (Oz up). The generalized coordinates are (r, θ), $\mathrm{CDOF} = 2$.
- **Hypotheses:** inertial frame S_0 with O at the hole; smooth hole and plane for parts (a)-(e); the cord stays taut; no drag.
- **Initial state:** $r(0) = a, \dot{r}(0) = 0, r(0)\dot{\theta}(0) = v_0$, hence $\dot{\theta}(0) = v_0/a$ and $\dot{z}_2(0) = 0$.

Symbol	Meaning	SI unit
m	mass of each particle	kg
a	half of the cord length	m
v_0	initial speed of P_1	m/s
g	gravitational acceleration	m/s^2
μ	friction coefficient (part f)	1

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, horizontal plane $z = 0$, $\mathbf{k}$ up.
- **Change of basis (polar basis of P_1).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

with $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$, $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$ (Poisson).

- **Kinematics:** $\mathbf{r}_1 = r\mathbf{e}_r$, $\mathbf{v}_1 = \dot r\mathbf{e}_r + r\dot\theta\mathbf{e}_\theta$; $\mathbf{r}_2 = (r-2a)\mathbf{k}$, $\mathbf{v}_2 = \dot r\mathbf{k}$.

$$T_{\text{kin}} = \tfrac{1}{2}m\left(\dot r^2 + r^2\dot\theta^2\right) + \tfrac{1}{2}m\dot r^2 = m\dot r^2 + \tfrac{1}{2}mr^2\dot\theta^2, \qquad V = mgz_2 = mg(r-2a)$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Circular motion, $r \equiv a$

> **Pedagogical Justification:** On the plane the only horizontal force on P_1 is the tension $-T\mathbf{e}_r$, which is central. For $r = a$ the radial acceleration is purely centripetal; P_2 must stay at rest.

$$P_1:\ m\left(\ddot r - r\dot\theta^2\right) = -T \xrightarrow{r=a,\ \ddot r=0} T = ma\dot\theta^2 = \frac{mv_0^2}{a}, \qquad P_2:\ m\ddot z_2 = T - mg = 0 \implies T = mg$$

$$\frac{mv_0^2}{a} = mg \implies v_0 = \sqrt{ag}$$

(b) Tension in the circular case

$$T = mg$$

(c) Quadratures for $v_0 = \sqrt{8ag/3}$

> **Pedagogical Justification:** The tension on P_1 is central, so $h = r^2\dot\theta$ is constant; the plane and hole are smooth and the cord inextensible, so $E = T_{\text{kin}} + V$ is constant. The two first integrals leave one equation for $r(t)$.

$$h = r^2\dot\theta = a v_0 = a\sqrt{\frac{8ag}{3}} \implies h^2 = \frac{8}{3}a^3 g$$

$$E = m\dot r^2 + \frac{mh^2}{2r^2} + mg(r-2a), \qquad E(0) = \tfrac{1}{2}m\cdot\frac{8ag}{3} - mga = \frac{1}{3}mga$$

Equating, dividing by m and isolating $\dot r^2$:

$$\dot r^2 + \frac{4a^3 g}{3r^2} + gr - 2ga = \frac{1}{3}ga \implies \dot r^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2} = \frac{g}{3r^2}\left(7ar^2 - 3r^3 - 4a^3\right)$$

Separating variables and applying Barrow's rule from $r(0) = a$:

$$\frac{dr}{dt} = \pm\sqrt{\frac{g}{3}}\,\frac{\sqrt{-3r^3 + 7ar^2 - 4a^3}}{r} \implies \int_a^r \frac{r'\,dr'}{\sqrt{-3r'^3 + 7ar'^2 - 4a^3}} = \pm\sqrt{\frac{g}{3}}\,t$$

(d) Oscillation between $r = a$ and $r = 2a$

> **Pedagogical Justification:** Motion requires $\dot r^2 \geq 0$, i.e. $P(r) = -3r^3 + 7ar^2 - 4a^3 \geq 0$. With $x = r/a$ the polynomial is $a^3(-3x^3 + 7x^2 - 4)$, and the hint gives its roots.

$$-3x^3 + 7x^2 - 4 = -(x-1)(x-2)(3x+2)$$

Check of the roots: $x = 1$: $-3 + 7 - 4 = 0$; $x = 2$: $-24 + 28 - 4 = 0$; $x = -2/3$: $\frac{8}{9} + \frac{28}{9} - 4 = 0$. Since $r > 0$, $3x + 2 > 0$ and $P \geq 0$ requires $(x-1)(x-2) \leq 0$:

$$1 \le x \le 2 \iff a \le r(t) \le 2a$$

The turning points ($\dot{r} = 0$) are $r_{\min} = a$ and $r_{\max} = 2a$, and the motion oscillates between them.

(e) Tension of the cord

> **Pedagogical Justification:** Apply Newton's second law to P_2 (weight and tension) with $\ddot{z}_2 = \ddot{r}$, because $z_2 = r - 2a$.

$$m\ddot{r} = T - mg \implies T = m\left(g + \ddot{r}\right)$$

Differentiate $\dot{r}^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2}$ with respect to time (chain rule, $\frac{d}{dt}r^{-2} = -2r^{-3}\dot{r}$):

$$2\dot{r}\ddot{r} = \left(-g + \frac{8a^3 g}{3r^3}\right)\dot{r} \implies \ddot{r} = -\frac{g}{2} + \frac{4a^3 g}{3r^3}$$

$$\boxed{T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)}$$

$T > 0$ for all $r \in [a, 2a]$, so the cord stays taut.

(f) Rough plane

Pedagogical Justification: Coulomb friction acts on P_1 opposite to its velocity, with magnitude μN_1. The normal force is $N_1 = mg$, because the cord is horizontal on the plane and only the weight acts vertically.

$$\mathbf{F}_f = -\mu mg\,\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = -\mu mg\,\frac{\dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta}{\sqrt{\dot r^2 + r^2\dot\theta^2}}$$

$$\mathbf{e}_r:\ m\left(\ddot r - r\dot\theta^2\right) = -T - \mu mg\frac{\dot r}{\sqrt{\dot r^2 + r^2\dot\theta^2}}$$

$$\mathbf{e}_\theta:\ m\left(r\ddot\theta + 2\dot r\dot\theta\right) = -\mu mg\frac{r\dot\theta}{\sqrt{\dot r^2 + r^2\dot\theta^2}}, \qquad P_2:\ m\ddot r = T - mg$$

Adding the $\mathbf{e}_r$ equation and the P_2 equation eliminates T; multiplying the $\mathbf{e}_\theta$ equation by r gives the evolution of h:

$$2\ddot r - r\dot\theta^2 = -g - \mu g\frac{\dot r}{\sqrt{\dot r^2 + r^2\dot\theta^2}}, \qquad \frac{d}{dt}\left(r^2\dot\theta\right) = -\mu g\,\frac{r^2\dot\theta}{\sqrt{\dot r^2 + r^2\dot\theta^2}}$$

Angular momentum and energy are no longer conserved (friction dissipates energy), so the problem cannot be reduced to the quadratures of (c).

PHASE 4

Physical Interpretation & Dimensional Verification

- **Tension at the turning points:** $T(a) = mg\left(\tfrac12 + \tfrac43\right) = \tfrac{11}{6}mg$ and $T(2a) = mg\left(\tfrac12 + \tfrac{4}{24}\right) = \tfrac23 mg$, both positive.
- **Consistency with (a):** for $v_0^2 = ag$, $h^2 = a^3 g$ and $\ddot r(a) = -\tfrac{g}{2} + \tfrac{h^2}{2a^3} = 0$, so $r = a$ is an equilibrium of the radial equation, as expected for the circular case.

- **Radial period** (consequence of (c) and (d)): with $x = r/a$ and $x = \frac{3}{2} - \frac{1}{2}\cos s$, $(x-1)(2-x) = \frac{1}{4}\sin^2 s$ and $dx = \frac{1}{2}\sin s\, ds$, so $\tau_r = 2\sqrt{\dfrac{3a}{g}}\displaystyle\int_0^{\pi} \frac{x(s)\,ds}{\sqrt{3x(s)+2}} \approx 6.342\sqrt{\frac{a}{g}}$.
- **Numerical check** (Python, RK4 on $\ddot r = -g/2 + 4a^3 g/(3r^3)$ with $a = g = 1, r(0) = 1$, $\dot r(0) = 0$): maximum radius 2.0000, period 6.3425 s (the quadrature gives 6.3424), and the energy-integral residual stays below 4×10^{-14}.
- **Dimensions:** $[T] = [m][g] = \mathrm{N}$ and $[v_0] = \sqrt{[a][g]} = \mathrm{m/s}$.
- **Key:** the official key (Problems.pdf) has no entry for Problem 43.

Source: Problems.pdf (PDF page 30); no key available